

# Math 300      Exam 3      Spring 2018

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Show all your work in the space provided for each problem. All problems will be graded for correctness, completeness, conciseness, and clarity. Do your best. Don't cut corners.

1. (10 pts) Use induction to prove that  $3n+1 < 3^n$  for all  $n \geq 2$

2. (6 pts) Prove by exhaustion that  $p(x) = x^2 + x$  has four zeros over  $\mathbf{Z}_{10}$ .

3. a) (4 pts) What does it mean **in terms of limits** to say the function  $f$  is continuous at a number  $a$  in its domain?

b) (4 pts) State carefully what this means **in terms of  $\varepsilon$  and  $\delta$** .

4. (10 pts) Use Cauchy's definition of limit to prove that  $\lim_{x \rightarrow 2} x^2 + x - 10 = -4$ .

5. a) (5 pts) What does it mean to say that  $a$  is algebraic over  $\mathbf{Q}$ ?

**Not on our exam**

b) (5 pts) Prove that  $3 + \sqrt{5}$  is algebraic over  $\mathbf{Q}$ .

**Not on our exam**

6. (8 pts) Let  $A$  and  $B$  be countably infinite sets that have no elements in common. Prove that the union of  $A$  and  $B$  will be countably infinite.

**Not on our exam.**

7. (12 pts) Prove that every linear function  $f(x) = mx + b$  with  $m \neq 0$  is one-to-one and find the inverse of the function.

8. (9 pts) Let  $G$  be the group  $U(7)$  under multiplication mod 7. Let  $f$  be the function from  $G$  to  $G$  that maps every element of  $G$  to its multiplicative inverse.

a) (6 pts) Fill in all missing values in the table.

|        |   |   |   |   |   |   |
|--------|---|---|---|---|---|---|
| $x$    | 1 | 2 | 3 | 4 | 5 | 6 |
| $f(x)$ |   |   |   |   |   |   |

b) (3 pts) Circle all answers that accurately complete the sentence:

The function  $f$  in part a is

an injection

a surjection

a bijection

a rejection

9. (10 pts) Let  $f$  be any function that is positive valued and differentiable at  $x = c$ .

a) According to the definition, what is the derivative of  $f$  at  $c$ ?

$$f'(c) =$$

b) Use the definition of the derivative to find the formula for the derivative of  $(f(x))^2$  at  $x = c$ .

10. (9 pts) Use one of Cantor's diagonalization arguments to prove that the set of all real numbers in the interval  $(0,1)$  is uncountable.

**Not on our exam**

11. (8 pts) Give examples of the following.

a) a real number that is not algebraic over  $\mathbf{Q}$

**Not on our exam**

b) a real number that is algebraic over  $\mathbf{Q}$  but is not rational

**Not on our exam**

c) a function from  $\{1,2,3,4\}$  to  $\{5,6,7\}$  that is a surjection

d) a function from  $\{1,2,3,4\}$  to  $\{5,6,7\}$  that is **not** a surjection